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Should we use single items? Better not☆

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ABSTRACT

Bergvist (2016, this issue) and Rossiter (2016, unpublished) claim that the conclusions of Sarstedt, Diamantopoulos, Salzberger, and Baumgartner (2016) are unwarranted as the study's methodology is flawed. This paper begs to differ.

1. Thank you, but…

Keeping with Oscar Wilde's famous quote “there is only one thing in life worse than being talked about, and that is not being talked about”, we very much appreciate the comments on our paper “Selecting single items to measure doubly concrete constructs: A cautionary tale” (Sarstedt, Diamantopoulos, Salzberger, & Wilczynski, 2016) by Bergkvist (in press) and Rossiter (2016). Considering that some of the paper's results question their research findings on the appropriateness of single-item measures (Bergkvist & Rossiter, 2007, 2009), coupled with the fact that both authors enjoy lively academic debates (e.g. Bergkvist, 2015, Rossiter, 2005, 2008), their rejoinders did not exactly come as a surprise. However, while considering the comments expressed therein, two questions emerged that made it very difficult to accept their criticisms. This response will share these two questions and some potential answers, hoping that they will help readers make up their own minds regarding single-item measurement and those elements of the C-OAR-SE procedure (Rossiter, 2002, 2011a, 2011b) that build on this measurement approach.

2. Where have all the expert raters (and users) of C-OAR-SE gone?

Construct definition in C-OAR-SE ultimately depends on rational expert judgment; or as Bergkvist (in press) notes: “the role of expert judges should be to determine the nature of the construct (…)”. For example, to classify a construct as doubly concrete, expert raters (or a majority thereof) have to agree that the construct has a simple, clear object coupled with a single and single-meaning attribute (Bergkvist, in press; Rossiter, 2002). If the expert raters agree on the doubly concrete nature of the construct, the use of a single item is—according to the C-OAR-SE procedure—sufficient (e.g. Bergkvist & Rossiter, 2007, Rossiter, 2002). Following Rossiter (2002, p. 310), these expert raters are a “small group of judges with expertise regarding the construct”. However, these experts must not have training in psychometrics as “it seems highly unlikely that experts in psychometrics would be able to provide an unbiased judgment of whether a construct could be measured with a single item” (Bergkvist, in press).

Two points are worth noting here. First, the potential benefits regarding single-item measurement and approaches to develop/select such measures have been around for decades (see Fuchs & Diamantopoulos, 2009 for a review of relevant literature) and certainly long before Bergkvist and Rossiter (2007, 2009) articles and Rossiter's (2002) C-OAR-SE procedure. Thus to claim that experts in psychometrics would be biased against single items is not only offensive but simply incorrect. Similarly, to somehow imply that C-OAR-SE is responsible for introducing the idea of single-item measurement to the literature is also incorrect. In fact, it was not in the context of C-OAR-SE but in psychometrics that single-item measurement was first discussed. Specifically, until the 1970s, the use of single items was generally accepted. The

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seminal works by Churchill (1979), Jacoby (1978), and Peter (1979) triggered a rethinking process that marked the start of multi-item use in marketing and business research in general. The widespread use of Confirmatory Factor Analysis further contributed to the advancement of psychometric adoption. Apart from that, the benefits and limitations of single-item use have long been discussed in a variety of disciplines outside marketing such as management (e.g., Kwon & Trail, 2005, Loo, 2002), psychology (e.g. Sackett & Larson, 1990, Wanous & Reichers, 1996), and various fields of medical research (e.g. Pomeroy, Clark, & Philp, 2001, Robins, Fein, Barton, & Green, 2001).

Second, in Sarstedt et al. (in press), the experts used were not only methodological experts but also had substantive knowledge/expertise in advertising and branding (a point ignored by Bergkvist, in press), which is exactly in keeping with the C-OAR-SE procedure. Moreover, there were thirteen experts that participated in the study. In comparison, all of Bergkvist and Rossiter's research (e.g. Bergkvist & Rossiter, 2007, 2009, Rossiter, 2002, 2008, 2009, 2011a, 2011b, 2012) relies on a single set of two expert raters: Lars Bergkvist and John R. Rossiter. This absence of variability in expert raters not only decreases confidence in the validity of the study findings but also may set the wrong standard if Rossiter's (2016, p. 6) prediction that "psychometrics in about 10 years' time will be found only in a museum" and will be substituted by C-OAR-SE becomes reality.

To date, however, the C-OAR-SE procedure has not been nearly as well received in applied research as the original paper's citation count and the winning of the 2012 *Jan-Benedict E.M. Steenkamp Award for Long Term Impact*, would suggest. To explore the dissemination of C-OAR-SE, we analyzed all articles in the Scopus database that have cited Rossiter's (2002) original C-OAR-SE article. The search in February 2016 yielded 645 citations from a variety of fields such as management, marketing, accounting, finance, and psychology. To narrow down the list of articles, the analysis focused on the field of marketing in which the C-OAR-SE procedure has been originally proposed and widely discussed (e.g. Diamantopoulos, 2005, Rigdon, Preacher, Lee, Howell, Franke, & Borsboom, 2011, Rossiter, 2005). More precisely, the review of articles in marketing consists of studies published in the top 20 marketing journals identified in Hult, Reimann, and Schilke's (2009) journal ranking, which related research also used (e.g. Hair, Sarstedt, Pieper, & Ringle, 2012, Hair, Sarstedt, Ringle, & Mena, 2012). The search yielded a total of 108 articles. Next, each article was read in detail and subsequently evaluated in terms of the context in which the Rossiter (2002) paper was cited. A senior graduate student and a professor proficient in the C-OAR-SE procedure independently developed a classification scheme based on an initial coding of 20 articles. The two classification schemes were then discussed and merged into a common one (see Table 1), which served as basis for the analysis of the remaining articles.

The results in Table 1 show that 38 of 108 articles (35.19%) cite Rossiter (2002) to justify the use of single items. 15 of these articles explicitly refer to the doubly concrete nature of the constructs under consideration (e.g., Fortenberry & McGoldrick, 2011), while the others generically indicate that the use of single items is appropriate in the

Table 1
Citation context of Rossiter (2002).

Context of citation	Number of articles (%)
Justify use of a single item	38 (35.19%)
Mention of C-OAR-SE as an approach to scale development	19 (17.59%)
Differentiate between formative and reflective measures	14 (12.96%)
Construct definition	12 (11.11%)
Emphasis on measures' content validity	10 (9.26%)
Comments and extensions of C-OAR-SE	8 (7.41%)
Application of the C-OAR-SE procedure	4 (3.70)
Use of specific answer scale formats	3 (2.78%)
Total	108 (100%)

context of their study (e.g., Patterson, Yu, & Kimpakorn, 2014). 19 of 108 articles (17.59%) referred to C-OAR-SE as an alternative approach to scale development (e.g., Ford, Mueller, & Taylor, 2011), sometimes indicating that they had considered Rossiter's (2002) criticism of standard scale development and testing procedures in their studies (e.g., Okazaki, Mueller, & Taylor, 2010). 14 articles (12.96%) cite Rossiter (2002) to highlight differences between reflective and formative measures and the need to carefully specify measurement models (e.g., Herington, Johnson, & Scott, 2009). 12 of the 108 articles (11.11%) cite the study to stress the need for careful construct definition (e.g., Nasco, Kulviwat, Kumar, & Bruner, 2008) or use (parts of) the C-OAR-SE procedure in this context (e.g., Brocato, Brocato, Voorhees, & Baker, 2012). Further contexts in which Rossiter (2002) has been cited include the need to consider content validity in measurement (9.26%), comments on the original and extensions of the C-OAR-SE procedure (7.41%), and the use of specific answer scale formats (2.78%). Most importantly, however, only 4 of 108 (3.70%) studies citing the original C-OAR-SE article *actually used the procedure to develop construct measures*. Cadeyux and Ng (2012) as well as Dickinger and Stangl (2013) use the C-OAR elements for construct measure validation and development, but do not use the procedure in full. In contrast, Sabri and Obermiller (2012) use C-OAR-SE to develop a measure of consumer perception of taboo in advertising but do not fully adhere to Rossiter's (2011b, p. 1585) suggestions as they rely on "faulty Likert answer scales". Finally, Rossiter (2012) introduces a contrastive measure that distinguishes brand love from brand liking, which focuses on content validity of the answer scale but does not make the application of the other aspects of the C-OAR-SE procedure transparent.

In short, the review shows that while many researchers *cite* C-OAR-SE, only very few actually *apply* the procedure (or even parts of it). Needless to say, this is not John R. Rossiter's fault and he frequently laments the fact that so few researchers have actually implemented his procedure (e.g., Rossiter, 2015). Nevertheless, we subscribe to Boshoff and Theron (2015, p. 264) who commented on the absence of C-OAR-SEicans by concluding that "as marketers this response should tell us something. A lot of people walked into the showroom, kicked the tires, but left without buying. Force feeding a market will not work. Fatigue is setting in. It's time to move on".

3. To predict or not predict: And if not, how?

The notion that single items of doubly concrete constructs exhibit at least the same predictive validity as their multi-item counterparts is one of the cornerstones of Rossiter's (2002, 2011a, 2011b) C-OAR-SE procedure and has been empirically tested by Bergkvist and Rossiter (2007, 2009). However, the relevance of the latter two studies for C-OAR-SE remains highly unclear. On the one hand, Bergkvist and Rossiter (2009, pp. 607–608) note that "Rossiter (2002) argues that single-item measures provide valid measurement of 'doubly concrete' constructs" and that "Bergkvist and Rossiter (2007) show that this argument holds empirically by demonstrating that the predictive validity of single-item measures of the doubly concrete constructs attitude towards the ad (A_{Ad}) and brand attitude (A_{Brand}) is equal to the predictive validity of multiple-item measures of the same constructs". On the other hand, Rossiter (2016, p. 5) emphasizes that "C-OAR-SE cannot be attacked by empirical, statistical arguments", which necessarily implies that empirical, statistical arguments cannot provide *support* for C-OAR-SE either. Bergkvist (2016, p. 2) notes that "the purpose of the Bergkvist and Rossiter studies was to empirically test claims made by psychometricians using tests favored by psychometricians (and the studies showed that those claims did not withstand empirical testing)". On the contrary, Rossiter (2016, p. 4) states that the only reason they considered predictive validity in their study titled "The predictive validity of multiple-item versus single-item measures of the same constructs", was "because the then-Editor of *JMR* insisted that we provide 'empirical proof'". Indeed, both authors seem to lack "unanimous agreement"

(Rossiter, 2002, p. 313) when it comes to the role of empirical validation for C-OAR-SE and the nature of their research.

Regardless of whether one interprets our study as "sniping at the C-OAR-SE approach" (Rossiter, 2016, p. 5), its results suggest that claims made by psychometricians using tests favored by psychometricians actually *do* withstand empirical testing: the findings clearly show that single items lag behind multi-item scales from a predictive validity perspective (see also Diamantopoulos, Sarstedt, Fuchs, Wilczynski, & Kaiser, 2012). But whence these differences in findings? Sarstedt and Wilczynski (2009) offer a potential explanation by pointing to some errors in the use of statistical tests. Specifically, in their analyses, Bergkvist and Rossiter's (2007, 2009) used Fisher's z transformation to test for significant differences between the predictor and criterion variables' correlations. However, this test should only be used when two *independent* correlations from two *different* samples are considered (e.g., Cohen, Cohen, West, & Aiken, 2002). The correlations in Bergkvist and Rossiter (2007, 2009) stem from a single sample and are therefore *not* independent, thus clearly violating the test's fundamental assumption that covariances between the correlations are zero. However, even the slightest violation of this assumption is highly problematic, making it difficult, or even impossible, to interpret the results in a meaningful way. When taking covariances into account, even small differences between correlations become significant and researchers should rely on paired-samples tests such as those by Meng, Rosenthal, and Rubin (1992) or Ferguson (1971), the latter of which is given by

$$t = \frac{(r_{12} - r_{13})\sqrt{(N-3) \times (1 + r_{23})}}{\sqrt{2 \times (1 - r_{12}^2 - r_{13}^2 - r_{23}^2 + 2 \times r_{12} \times r_{13} \times r_{23})}}. \quad (1)$$

In this formula, r_{12} is the correlation between predictor and criterion when both variables are measured with multiple items, r_{13} is the correlation between predictor and criterion when either is measured with a single item, and r_{23} is the correlation between those variables that are not common to either model. For example, when comparing the predictive validity of single- and multi-item measures of A_{Ad} with regard to A_{Brand}, r_{23} is the correlation between the single- and multi-item measures of A_{Ad}. Naturally, this correlation is going to be very high, particularly when the single item stems from a homogeneous scale with high internal consistency.

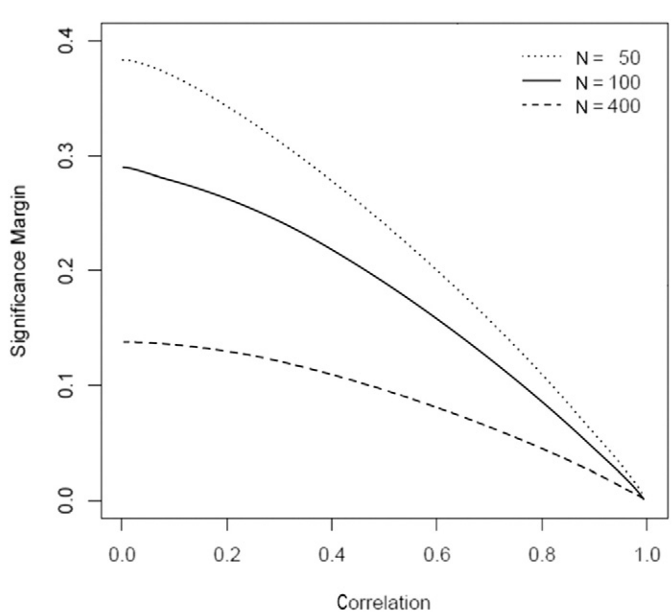

Fig. 1. Significance margins for Fisher's z test.

To illustrate the consequences of this test violation, consider Figs. 1 and 2. Using the correlation between single and multi-item r_{13} as a benchmark, Fig. 1 indicates how great the difference between r_{12} and r_{13} would have to be in order to render this difference significant at 5% when using Fisher's z test. For example, with $N = 100$ and assuming a correlation between a single-item measure and the multi-item measure of 0.6 (as in many cases in Bergkvist & Rossiter, 2009), the correlation between two multi-item measures would have to be 0.15 units higher (or lower) to indicate a significant difference between the single and multi-item measurement. Even with heterogeneous construct measures exhibiting low levels of internal consistency, such high differences between single- and multi-item measures are highly unlikely to occur. However, when (correctly) applying Ferguson's (1971) test and assuming a correlation r_{23} between the single and multi-item predictor of 0.85, as observed in the vast majority of corresponding comparisons in our study, the difference in correlations would have to be 0.075—half of that when (incorrectly) applying Fisher's z test. For higher levels of r_{23} (e.g., 0.95) and higher sample size (e.g., 400), the significance margin drops to 0.02, which would render the (very small) differences between single- and multi-item measures in Bergkvist and Rossiter (2007) significant.

These results clearly indicate that the violation of the test requirements has dramatic consequences for the results, particularly when considering the differences in correlations between single and multi items as reported in Bergkvist and Rossiter (2009). For example, in their contrasting of single- and multi-item measures of A_{Ad} to predict single- and multi-item measures of A_{Brand}, the multi-item measures yielded higher correlations in 29 of 40 comparisons (72.5%) with an average difference of 0.17 and range in differences between 0.01 and 0.39.[1]

Against this background, it is not surprising that Diamantopoulos et al.'s (2012) simulation study on the influence of design characteristics (e.g., the average inter-item correlation among the items of the predictor and criterion constructs, the number of items used to measure these constructs) on the predictive validity of single- versus multi-item measures shows that, under most conditions typically encountered in practical applications, multi-item scales clearly outperform single items in terms of predictive validity. The results show that single-item measures should be considered only in situations when working with small sample sizes (i.e., $N < 50$), *and* the expected effect sizes are small, *and* the items of the originating multi-item scale are highly homogeneous (i.e., Cronbach's alpha > 0.90), *and* the items are semantically redundant (see also Kamakura, 2015). In principle, the suitability of a single-item vis-à-vis a multi-item scale can even be assessed analytically (Rozeboom, 1979). To make matters worse, even if one knowingly accepts that single-item measures are likely to lag behind multi items, identifying an appropriate item by means of rater assessments or statistical criteria is highly problematic (Sarstedt et al., in press).

This situation is quite alarming, considering (1) that the majority of studies (35.19%) in the above review have cited Rossiter's (2002) original C-OAR-SE article to justify their use of single items, and (2) the popularity of Bergkvist and Rossiter's (2007) study as evidenced in more than 600 citations in the Scopus database as of February 2016. Independent of these findings' implications for C-OAR-SE, researchers should be aware that the use of single items triggers type II errors, thereby having adverse consequences for hypothesis testing and hence theory-building.

[1] Note that Bergkvist and Rossiter (2009) do not report significances for the differences between multi and single items of the same scale but only for the differences between the multi-item scale and a global single item as well as between the global single item and each individual scale item. This assessment does not allow making conclusions about any differences between the multi-item scale and each individual scale item.

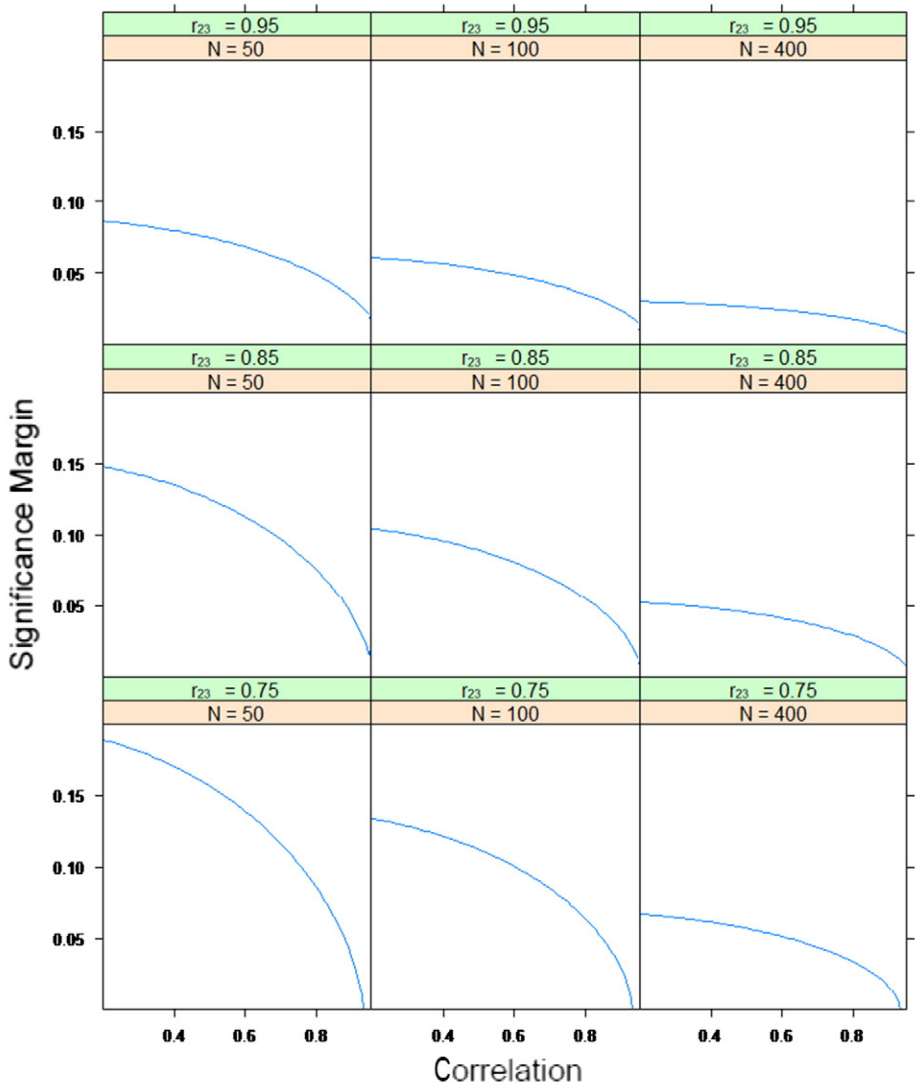

Fig. 2. Significance margins for Ferguson's (1971) test.

4. Conclusion

We are thankful for Bergkvist's (in press) and Rossiter's (2016) comments but for reasons depicted above are hesitant to accept their criticisms. In fact, their rejoinder misses the central point of the debate. There are many appealing reasons to use single items (e.g., Fuchs & Diamantopoulos, 2009) but predictive validity is certainly not one of them. Regardless of whether a construct is a doubly concrete construct, a concrete construct, or even just a construct, single items lag behind multi-item scales in terms of predictive validity—there is no need to go into a deep philosophical or theoretical discussion to justify their use. This is an empirical issue and has nothing to do with any underlying conceptual considerations in C-OAR-SE or any other scale development approach.

Should researchers therefore never use a single-item measure? Absolutely not! Sometimes, the study setting necessitates the use of single items, for example, when the survey length or questionnaire format is restricted due to budget constraints, difficulties in recruiting respondents, limited population size, or the need to collect dyadic data (Fuchs & Diamantopoulos, 2009), among other reasons. However, when using single items, researchers must accept the fact that single items have lower predictive power, which can easily trigger type II errors. Since null results face a higher barrier to publication than those that yield statistically significant differences (e.g., Franco, Malhotra, & Simonovits, 2014), the use of single items can easily backfire on researchers. Yes, this is unfair, but such is life.

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